End of Semester II/Res-sit Examination Solved Summary: Feb, 2022

Course Code: CSS 114

Course Title: Databases and Database Systems

Programme: BSc. ITS I, BSc. ICTB I & BSc. EDU(MICT) I

INSTRUCTIONS

1. There are **FOUR (4)** question in this paper
2. Answer **ALL** Questions

QUESTION ONE (25 MARKS)

a) What is a relation? **(4 Marks)**

Possible answer:

A Relation is a two-dimensional table consists of rows (records) and columns (attributes or fields).

b) Use data in relation PROJECT below to demonstrate three types of anomalies that can arise in a relation **(21 Marks)**

PROJECT

Project Code	Project Title	Project Manager	Project Budget	Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
PC010	Pensions System	M Phillips	24500	S10001	A Smith	L004	IT	22.00
PC010	Pensions System	M Phillips	24500	S10030	L Jones	L023	Pensions	18.50
PC010	Pensions System	M Phillips	24500	S21010	P Lewis	L004	IT	21.00
PC045	Salaries System	H Martin	17400	S10010	B Jones	L004	IT	21.75
PC045	Salaries System	H Martin	17400	S10001	A Smith	L004	IT	18.00
PC045	Salaries System	H Martin	17400	S31002	T Gilbert	L028	Database	25.50
PC045	Salaries System	H Martin	17400	S13210	W Richards	L008	Salary	17.00
PC064	HR System	K Lewis	12250	S31002	T Gilbert	L028	Database	23.25
PC064	HR System	K Lewis	12250	S21010	P Lewis	L004	IT	17.50
PC064	HR System	K Lewis	12250	S10034	B James	L009	HR	18.50

Possible answer:

i. **Insertion Anomaly** – *adding new rows forces user to create duplicate data.*

From the relation PROJECT

If new employee is employed, for project PC010, PC045 0r PC064 of the same department, department details and Project details must be re-entered causing duplication.

ii. **Deletion Anomaly** – *deleting rows may cause a loss of data that would be needed for other future rows*

From the relation PROJECT

- If we remove **Employee S10030 (L Jones),** we lose the information about the existence of **Pensions department.**
- Again, if we remove an **Employee S10034 (B James)** we lose the information about the existence of **HR department.**

iii. **Modification (Update) Anomaly** – *changing data in a row force changes to other rows because of duplication.*

From the relation PROJECT

- Changing an employee department such as Employee S10001 and S21010, forces us to update multiple records.
- Also, changing the Project manager for the Project PC010, PC045 and PC064, forces us to update multiple records too.

QUESTION TWO (20 MARKS)

State whether each of the following assertions is true **(use T)** or false **(use F).** **(@2 marks)**

a) A relation that has no partial dependencies is in 2NF **F**

b) A relation in 2NF has the following anomalies: insertion, deletion, and update. **T**

c) A relation is an association between two or more tables in E-R model. **F**

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d) A table with repeating groups has to be broken down into two to remove the repeating groups. **F**

e) A composite key is a primary key that has two attributes. **F**

f) The entity integrity rule states that either each foreign key value must match a primary key value in another relation or the foreign key value must be null. **F**

g) We use a surrogate key as a means to deal with complicated primary keys. **T**

h) "DROP TABLE Student;" deletes all data in the student table. **F**

i) The appropriate data type for a person's phone number of Integer **F**

j) One advantage of the database approach is to eliminate redundancy of data. **F**

QUESTION THREE **(20 MARKS)**

a) What is normalization? In which stage of database design is it done? Explain whether it is necessary step? **(10 Marks)**

Possible answer:

- ***Normalization*** *is the process of decomposing relations with anomalies to produce smaller,* ***well-structured*** *relations with less anomalies.*
- Normalization is performed in a **Logical Design** stage.
- Normalization is not a necessary stage; it is implemented so that to reduce anomalies but the database can still run without normalization although will not run that much smoothly.

b) Consider an E-R model with entity type LECTURER and COURSE. Considering MU's business rules:

Is it possible to end up with two relations when this model is converted to a logical model? Explain. **(10 Marks)**

Possible answer:

NO.

MU's business rules:

- One LECTURER may teach many COURSES.
- One COURSE must be taught by many LECTURERS

This gives us a Many-to-many relationship.

From that relationship, many to many will end-up with three relations because when mapping E-RD into relation we need to create a new relation with the primary keys of the two entities as its primary key.

QUESTION FOUR **(35 MARKS)**

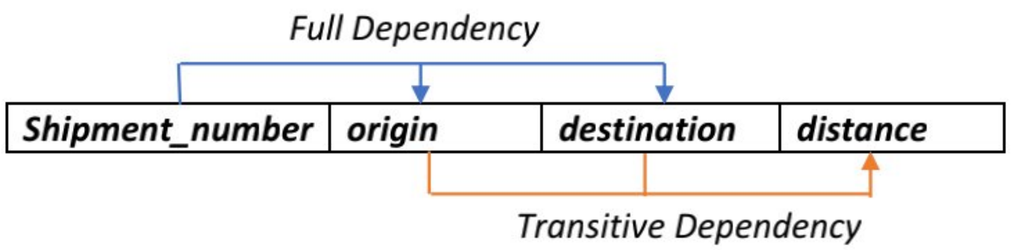
Table shipment presents a portion of shipment data for a large logistics company. Each shipment (identified by shipment number) uniquely identifies the shipment origin, destination, and distance (Km). the shipment origin and destination pair also uniquely identify the distance.

Shipment.

Shipment_number	***origin***	***destination***	***distance***
409	Dar es Salaam	Lusaka	1941
618	Mwanza	Nairobi	680
723	Dar es Salaam	Kigali	1424
824	Tunduma	Lubumbashi	915
629	Mtwara	Maputo	3066
319	Arusha	Mamanga	109

a) Convert Shipment to the next normal form and show the resulting table(s) with the sample data presented in Shipment. **(10 Marks)**

Possible answer:



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Route

origin	*destination*	*distance*
Dar es Salaam	Lusaka	1941
Mwanza	Nairobi	680
Dar es Salaam	Kigali	1424
Tunduma	Lubumbashi	915
Mtwara	Maputo	3066
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Use your answer for part a) to answer the following questions.

b) Write a query to display all shipments but those whose origin is Dar es salaam. **(5 Marks)**

Possible answer:

```
select * from shipment where origin = 'Dar es salaam';
```

c) Write a query to display shipment# and distance for each shipment. **(5 Marks)**

Possible answer:

```
select Shipment.shipment_number, Route.distance FROM
shipment JOIN Route on Route.origin = shipment.origin and
Route.destination = shipment.destination;
```

d) Write a query to correct the value for destination for route 319 to read "Namanga". **(5 Marks)**

Possible answer:

```
UPDATE shipment SET destination = 'Namanga'
WHERE shipment_number = 317;
```

e) A new shipment (#755) from Dar es salaam to Unguja (distance currently unknown) has been introduced. Update the database with this information. **(5 Marks)**

possible answer:

```
insert into Route (Origin, destination) values ('Dar es
salaam','Unguja');

insert into shipment values (755,'Dar es salaam',
'Unguja');
```

f) Shipment #629 has to be shut down. Write a query to remove the route information from the database **(5 Marks)**

Possible answer:

```
DELETE from shipment WHERE shipment_number =
629;
```


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